

Chapter - 1.

Electric charges & Fields.

Intro :-

Electrostatic

Charges

rest.

Electrostatics deals with the study of forces, fields and potentials arising from static charges.

Way of charging :-

- 1) friction $\leftarrow$ Rubbing
- 2) Induction $\leftarrow$ touching
- 3) Conduction $\leftarrow$

Glass Rod $\rightarrow$ +ve

Plastic Rod $\rightarrow$ -ve

Fur $\rightarrow$ +ve

Silk $\rightarrow$ -ve

$$1 \text{ electron} = -1.6 \times 10^{-19} \text{ C}$$

$$1 \text{ Coulomb} = 6.25 \times 10^{18} \text{ electron of charge}$$

Electric charge

It is the property of the matter which gives the ability to repel (or) attract other charged substance placed in its vicinity.

Properties of Electric Charge

- Additivity of charges.
- Quantization of charges.
- Conservation of charges.

Additivity of charges

Total charge of the system is obtained by adding all the charges with their sign..

- If a system contain n charges. Then ;

$$q_1 + q_2 + q_3 + \dots + q_n$$

$\boxed{= +2e}$

$\frac{C}{\rightarrow}$

- Charges has magnitude only. Similar to mass, No direction.

Quantization of charge

The smallest charge that can be added (+) or removed from the body is the charge of an electron (e).

If charge on a body is Q , then

$$Q = ne$$

Where $n =$ Integers.

$$e = -1.6 \times 10^{-19} \text{ C}$$

..... $(-3, -2, -1, 0, 1, 2, 3)$

$\Rightarrow$ Charge is never be fraction.

like : $+1.5e$, $-2.6e$
 $\times$ $\times$

Conservation of charges

For an isolated system, charge can neither be created nor be destroyed i.e., net charge of an isolated system remains constant.

Insulator can be charged by friction.

eg: glass attract paper.

Properties of conductor :-

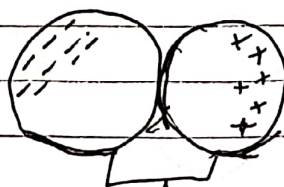
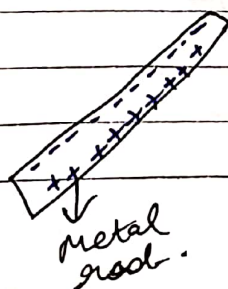
- Two conductors kept in contact behave as one.
- The charge given to a conductor distributes itself on the outer surface of the conductor only. The ~~dis~~ distribution may (or) may not be uniform. This is because the way you charge the object.
- There are 2 most common methods of charging of a conductor.

i) conduction. [touching]

ii) Induction

Induction :

The process of redistribution of charges on the surface of conductor when another charged object is brought near the conductor, is called induction.

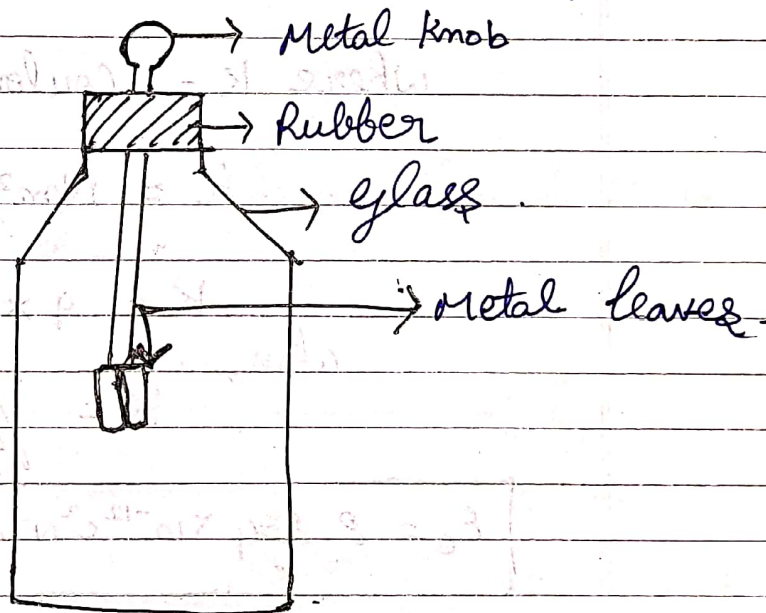


Total charge of the Metal ball is 0

because charge is neither be create nor be destroy for an isolated system.

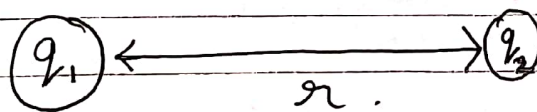
Detection of charge on object.

Charge on an object can be detected by using a device called electroscope.

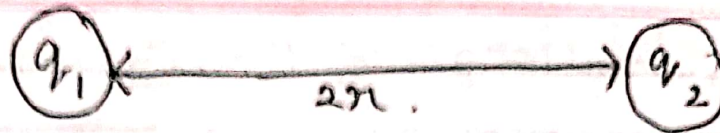


=> gold leaves electroscope.

Force between two multiple charges:-



$$F \propto q_1 q_2$$



$$F \propto \frac{1}{r^2}$$

$$F \propto \frac{q_1 q_2}{r^2}$$

$$F = K \frac{q_1 q_2}{r^2}$$

~~Coulomb's Law~~

Coulomb's Law

where K = Coulomb's Constant

$$K = \text{Nm}^2/\text{C}^2$$

$$K = 9 \times 10^9 \text{ [in vacuum]}$$

also,

$$K = \frac{1}{4\pi\epsilon_0}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

→ "epsilon" permittivity of free space

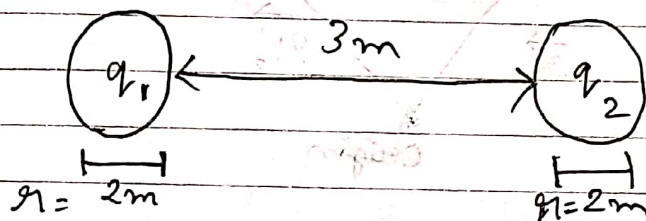
$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2}$$

//

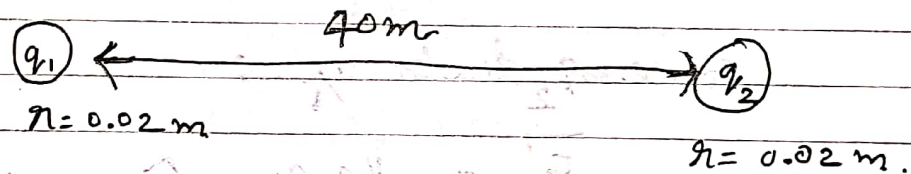
Limitations of "Coulomb's Law !"

$\Rightarrow$ Coulomb's law is not applicable as distance between charged spheres and radius is comparable.

radius $\ll$ distance.



$\Rightarrow$ here This Law is not applicable.



$\Rightarrow$ here The Law is applicable.

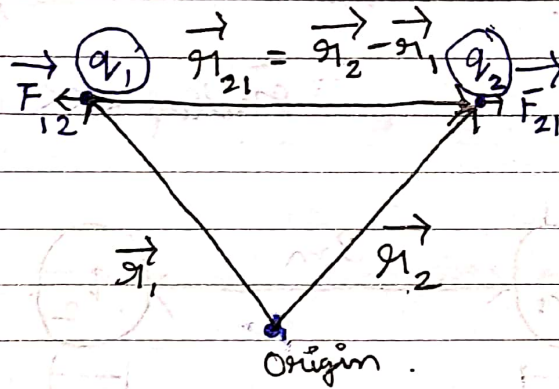
$\Rightarrow$ The charges to be considered as Point charges.

Limitation of coulomb's law.
Overview:-

- i) $r \ll d$
- ii) point size charge.
- iii) charges must be rest.

Vector form of Coulomb's Law

Let consider The q_1 and q_2 be +ve charge.



$$|F| = \frac{k q_1 q_2}{|\vec{r}_2 - \vec{r}_1|^2}$$

Force on q_2 due to q_1 :-

$$\vec{F}_{21} = \frac{k q_1 q_2}{|\vec{r}_2 - \vec{r}_1|} (\hat{r}_{21}) = \frac{k q_1 q_2}{r_{21}^2} \hat{r}_{21}$$

Force on q_1 due to q_2 :-

$$\vec{F}_{12} = \frac{k q_2 q_1}{r_{21}^2} (-\hat{r}_{21})$$

$$\boxed{\vec{F}_{21} = -\vec{F}_{12}}$$

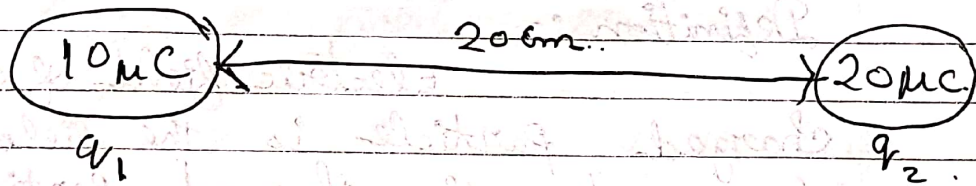
Ques:

Q) Two charges of $10\mu\text{C}$ and $20\mu\text{C}$ are separated by 20cm . The ratio of force acting on them will be.

a) 1:1

b) 1:2

c) 1:4



$$\frac{|\vec{F}_{12}|}{|\vec{F}_{21}|} = \text{ratio} = ?$$

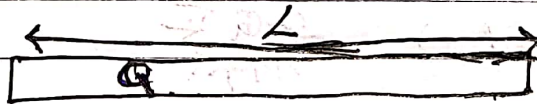
W.K.T:-

$$\vec{F}_{12} = -\vec{F}_{21}$$

a) 1:1

Continuous charge distribution

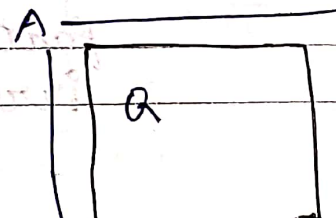
i) linear charge distribution.



$$\lambda = \frac{Q}{L}$$

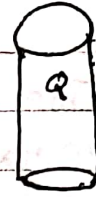
ii) surface charge distribution.

$$\sigma = \frac{Q}{\text{Area}}$$



iii) Volume charge distribution.

$$\rho = \frac{Q}{\text{Vol.}}$$

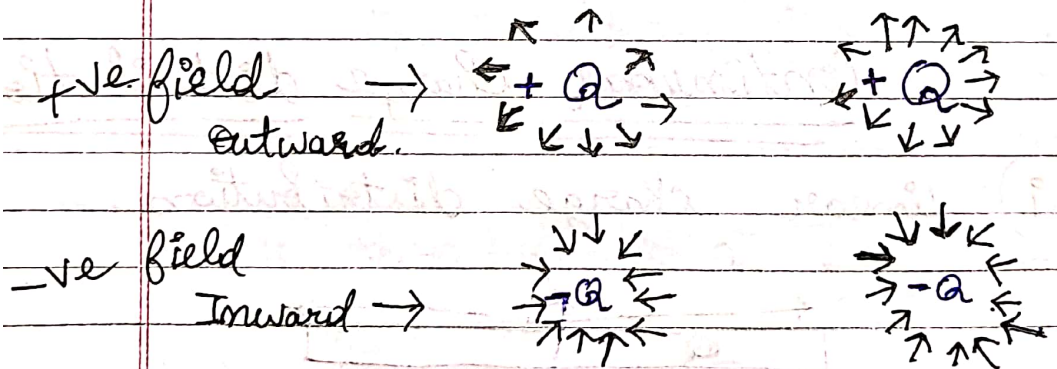


Electric Field

Definition :-

Electric field of a charged particle is the field surrounding the charged particle in which it can exert an influence on another charged particle.

⇒ "It Influence the presence the charged particle."



Properties of Electric field :-

- 1) Electric field of every charged particle extends upto infinity irrespective of the magnitude of

charge.

- 2) The magnitude (strength) of an electric field can be measured by its effect on charges located in the field.

a) Where the force is greatest on the test charges, the field is strongest.

b) where the force on the test charges is weak, the field is small.

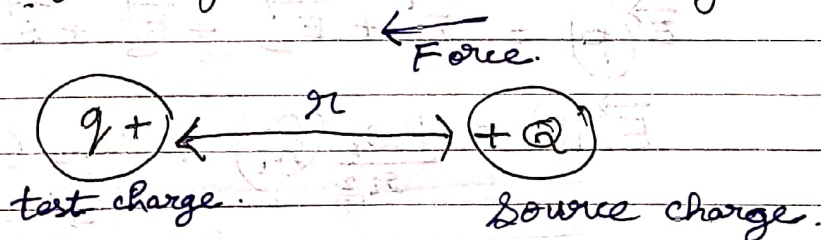
Formula :-

$$\vec{E} = \frac{\vec{F}}{q_0}$$

$\vec{E}$ = electric field.

$\vec{F}$ = Force exerted by the test charge.

q_0 = Magnitude of test charge.



$$F = \frac{kqQ}{r^2}$$

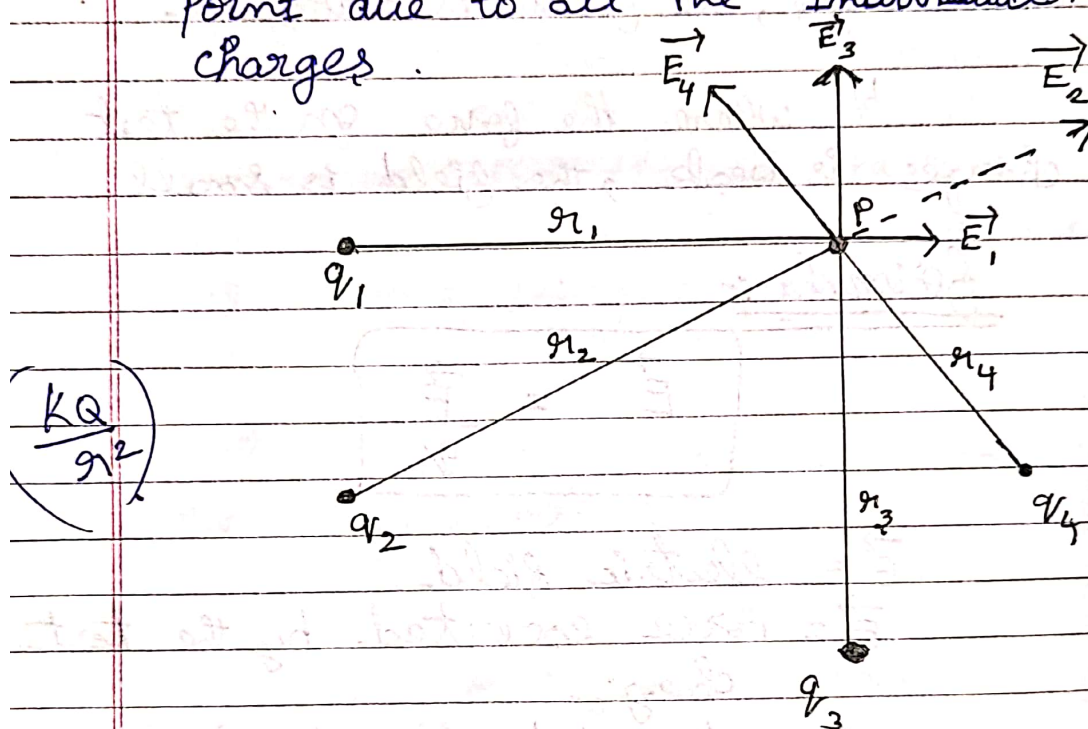
$$\vec{E} = \frac{\vec{F}}{q_0} \Rightarrow \vec{E} = \frac{kQq_0}{q_0 r^2} (\hat{r}) \Rightarrow \boxed{\vec{E} = \frac{kQ}{r^2} (\hat{r})}$$

Note: Electric field is only depend on source charge.

SI unit of $\vec{E} = \text{N/C} = \frac{\text{Force}}{\text{charge}}$

Electric field due to system of charges.

Electric field at a point due to a system of charges is the vector sum of the electric field at the point due to all the individual charges.



$$\vec{E}(P) = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4$$

$$\vec{E}(P) = \frac{kQ_1}{r_{1P}^2} (\hat{r}_1) + \frac{kQ_2}{r_{2P}^2} (\hat{r}_2) + \frac{kQ_3}{r_{3P}^2} (\hat{r}_3) + \frac{kQ_4}{r_{4P}^2} (\hat{r}_4)$$

$$\vec{E}(P) = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r_{1P}^2} \hat{r}_1 + \frac{q_2}{r_{2P}^2} \hat{r}_2 + \frac{q_3}{r_{3P}^2} \hat{r}_3 + \frac{q_4}{r_{4P}^2} \hat{r}_4 \right)$$

$$\frac{q_3}{r_{3P}^2} \hat{r}_3 + \frac{q_4}{r_{4P}^2} \hat{r}_4$$

Ques.

Q) Two charges of $1.0 \mu\text{C}$ and $2.0 \mu\text{C}$ are placed 10 cm apart. At what distance (in cm) from the $1 \mu\text{C}$ charge (on the line joining the two charges) is the electric field zero?

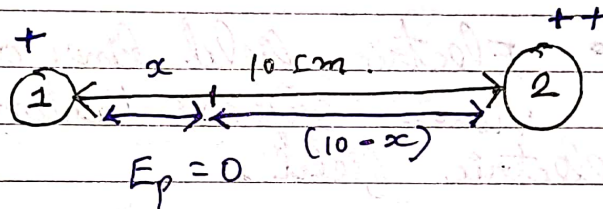
a) 5.86 cm

c) 3.33 cm

b) 4.14 cm

d) 6.66 cm

Sol:



$$\Rightarrow \frac{K \cdot 1 \mu\text{C}}{x^2} \cdot \hat{i} + \frac{K \cdot 2 \mu\text{C}}{(10-x)^2} \cdot (-\hat{i}) = 0$$

$$\Rightarrow \frac{1 \mu\text{C}}{x^2} = \frac{2 \mu\text{C}}{(10-x)^2}$$

$$\Rightarrow \left(\frac{10-x}{x} \right)^2 = 2$$

$$\Rightarrow \frac{10-x}{x} = \sqrt{2}$$

$$\Rightarrow 10-x = \sqrt{2}x$$

$$\Rightarrow 10 = (\sqrt{2}+1)x$$

$$\Rightarrow \frac{10}{\sqrt{2}+1} = x$$

$$x = \frac{10}{1.414+1} = \frac{10}{2.414}$$

$$x = 4.14 \text{ cm}$$

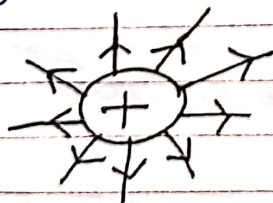
option (b).

∴ Properties Electric lines of forces..

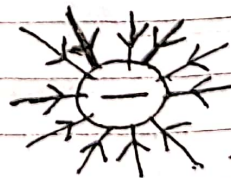
- Electric field lines are a visual representation of the net electric field.

- It is a curve drawn in such a way that the tangent to it at each point is in the direction of the net field at that point.

- A field line is purely a geometric representation and has no physical existence.



Outward

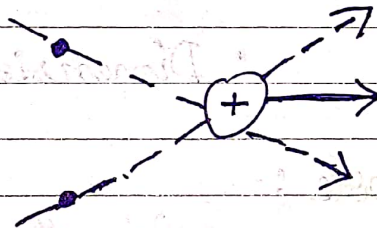


Inward.

- Electric field lines always represent the net electric field at a point.

- Electric field lines can't intersect each other.

because; If electric field lines intersect then there will be two tangent and two direction for one charge. This is not possible...

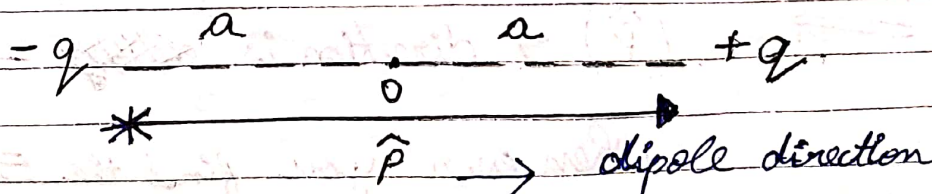


Not possible.

- If electric field ~~lines~~ density is high at a point then the no. of electric field line will be higher. and vice versa.

Electric dipole.

It defined as the simple product of magnitude of either charge and the distance of separation between the two charges and it is denoted by



$$\vec{P} = q(2a)\hat{P}$$

- It is a vector quantity.
- Dipole moment always directed from $-q$ to $+q$.

- SI unit: ~~Coulomb~~ coulomb metre

$$I = q/t$$

$$q = It$$

$$(C \cdot m)$$

$$\downarrow$$

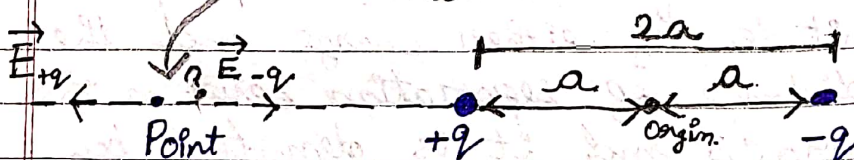
$$(C \cdot m)$$

- Dimension: $[M^0 L^1 T^1 A^1]$

Point dipole :-

When the dipole distance $(2a)$ approaches zero, the charge q approaches infinity in such a way that the product $P = q(2a)$ is finite. Such a dipole is referred as a point dipole.

Electric field due to dipole at axial line.



Note:

- $(\hat{P}) \rightarrow$ direction is ~~always~~ $-ve$ to $+ve$.

- When ever you find the Electric field at a point, assume there is a $+ve$ point charge at the

Derivation:-

PAGE No

DATE: / /

$$\vec{E}_p = \frac{Kq}{(r-a)^2} (-\hat{i}) + \frac{Kq}{(r+a)^2} (+\hat{i})$$

$$\vec{E}_p = \frac{Kq}{(r-a)^2} (\hat{p}) + \frac{Kq}{(r+a)^2} (-\hat{p})$$

Take $\hat{p}$ as common:-

$$\vec{E}_p \Rightarrow \hat{p} \left(\frac{Kq}{(r-a)^2} - \frac{Kq}{(r+a)^2} \right)$$

$$\vec{E}_p = Kq \hat{p} \left(\frac{4ar}{(r^2 - a^2)^2} \right)$$

How

$$\vec{E}_p = Kq \hat{p} \left(\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2} \right)$$

$$\vec{E}_p = Kq \hat{p} \left(\frac{(r+a)^2 - (r-a)^2}{(r-a)^2 (r+a)^2} \right)$$

$$(a+b)^2 - (a-b)^2 = 4ab$$

Final :

$$\vec{E}_p = Kq \hat{p} \left(\frac{4ar}{(r^2 - a^2)^2} \right)$$

$$P = (2a)q$$

$$\vec{E}_p = \frac{2(K) \hat{p} P r}{(r^2 - a^2)^2} = \frac{2K \hat{p} P r}{(r^2 - a^2)^2}$$

Note : $\boxed{p \times \hat{r} = \vec{p}}$

PAGE No.

DATE

$$\vec{E}_p = \frac{2K \vec{p} r}{(r^2 - a^2)^2}$$

$$\Rightarrow r \gg a,$$

$$\vec{E}_p = \frac{2K \vec{p} r}{(r^2)^2} = \frac{2K \vec{p} r}{r^4}$$

$$\boxed{\vec{E}_p = \frac{2K \vec{p}}{r^3}}$$

Quiz

- q.) Two charges each of magnitude $1 \mu\text{C}$ but opposite in sign are 1 cm apart. Calculate the magnitude of electric field at a point distance 10 cm from the mid point on axial line of the dipole.

Sol

Given :-

$$K = 9 \times 10^9 \rightarrow \text{constant}$$

$$r = 10 \text{ cm}$$

$$2a = 1 \text{ cm}$$

$$q = 1 \mu\text{C}$$

$$\text{dipole } p = 2aq$$

$$p = 1 \text{ cm} \times 1 \mu\text{C} = \frac{1}{100} \times 10^{-6}$$

formula :-

$$\vec{E}_p = \frac{2 \times 9 \times 10^9 \times \frac{1}{100} \times 10^{-6}}{(10)^3}$$



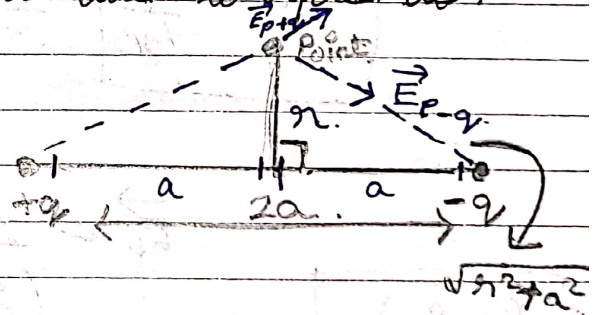
$$\vec{E}_p = \frac{18 \times 10^9 \times 0.01 \times 10^{-6}}{\frac{10^3}{10^3}}$$

$$\vec{E}_p = \frac{18 \times 10^9 \times \frac{1}{100} \times 10^{-6}}{\frac{1}{10^3}}$$

$$\begin{aligned} \vec{E}_p &= 18 \times 10^9 \times 10^{-6} \times 10^3 \times \frac{1}{100} \\ &= 18 \times 10^6 \times 0.01. \end{aligned}$$

$$\boxed{\vec{E}_p = 1.8 \times 10^5 \text{ N/C}}$$

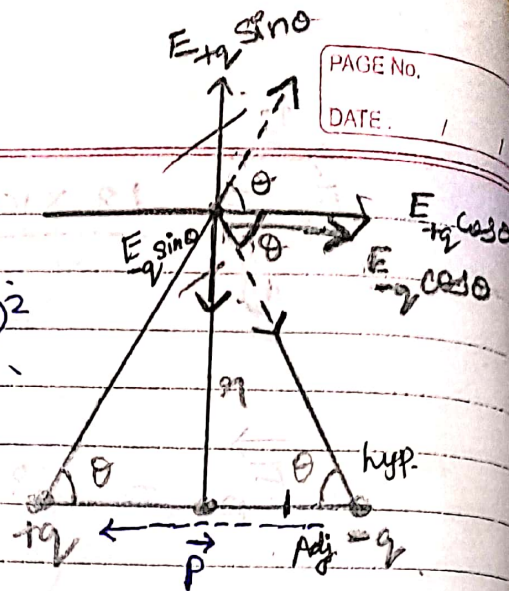
Electric field due to dipole at equatorial position.



$$\vec{E}_p = \vec{E}_{-q} + \vec{E}_{+q}$$

$$\vec{E}_p = \frac{kq}{(\sqrt{r^2 + a^2})^2} \hat{r}_{p \rightarrow -q} + \frac{kq}{(\sqrt{r^2 + a^2})^2} \hat{r}_{p \rightarrow +q}$$

$$\vec{E}_P = \frac{kq}{(\sqrt{x^2+a^2})^2} + \frac{kq}{(\sqrt{x^2+a^2})^2}$$



Resultant :- $E_+ \cos \theta$ & $E_- \cos \theta$

$$\vec{E}_P = 2E \cos \theta (-\hat{P})$$

$$\vec{E}_P = \frac{2Kq}{(x^2+a^2)} \cos \theta (-\hat{P})$$

$$\boxed{\cos \theta = \frac{\text{Adj.}}{\text{hyp.}}} \Rightarrow \cos \theta = \frac{a}{\sqrt{x^2+a^2}}$$

$$\vec{E}_P = \frac{2Kq}{(x^2+a^2)} \frac{a}{\sqrt{x^2+a^2}} (-\hat{P})$$

$$\boxed{P = (2a)q}$$

$$\boxed{P \times \hat{P} = \vec{P}}$$

$$\vec{E}_P = \frac{K \cdot P \cdot (-\hat{P})}{(x^2+a^2)^{1/2} (x^2+a^2)^{1/2}}$$

$$\boxed{\vec{E}_P = \frac{-K \vec{P}}{(x^2+a^2)^{3/2}}}$$

formula :-

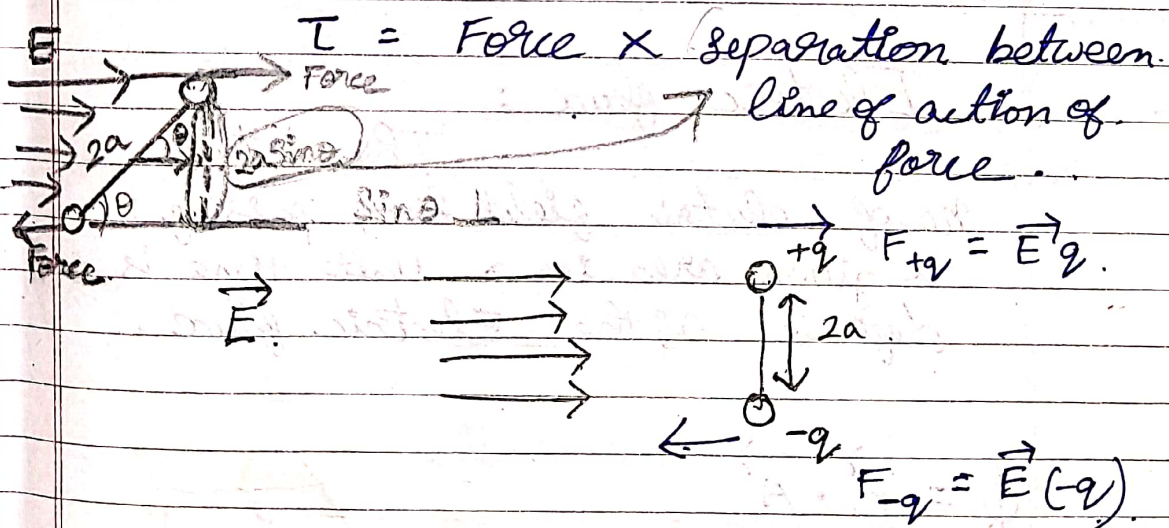
$$1) \text{ Axial} = \frac{2K\vec{P}}{r^3}$$

$$2) \text{ Equatorial} = \frac{-K\vec{P}}{r^3}$$

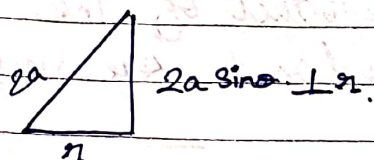
Dipole in a uniform external electric field :-

Torque :

A equal and opposite force act with a separation in their line of action, they produce a couple. The torque due to this couple.



Net force is zero in uniform external electric field.



$$\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{r}{2a} = \sin \theta \rightarrow \boxed{2a \sin \theta = r}$$

$$p = (2a)q$$

$$\tau = (qE) \times 2a \sin \theta$$

$$\tau = PE \sin \theta$$

In vector form :-

$$\tau = \vec{p} \times \vec{E}$$

Here the direction of Torque into the plane...

formula:

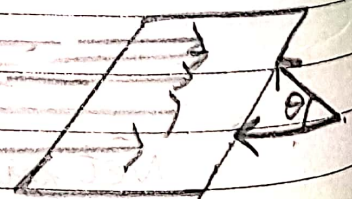
1) $|\vec{\tau}|$ magnitude = $PE \sin \theta$

2) $\vec{\tau} = \vec{p} \times \vec{E}$

Electric flux:

The total no. of electric field lines passing a given area in a unit time is defined as the Electric flux.

$$\phi = \vec{E} \cdot \vec{A}$$



- It is the dot product of the Electric field and Area vector.

formula: To Find flux.

$$\phi = |\vec{E}| |\vec{A}| \cos \theta$$

- flux is ~~zero~~ ^{zero} when $\theta = 90^\circ$.

$\Rightarrow$ because the Area is $\perp$ to Electric field line.

Gauss's law

The total charge enclosed in a closed surface is proportional to the total flux enclosed by the surface. There, if ϕ is total flux and ϵ_0 is electric constant, the total electric charge Q enclosed by the surface is.

$$Q \propto \phi$$

$$Q = \phi \epsilon_0$$

$$\phi = \frac{Q}{\epsilon_0}$$

where,

$Q \rightarrow$ electric charge

$\phi \rightarrow$ flux

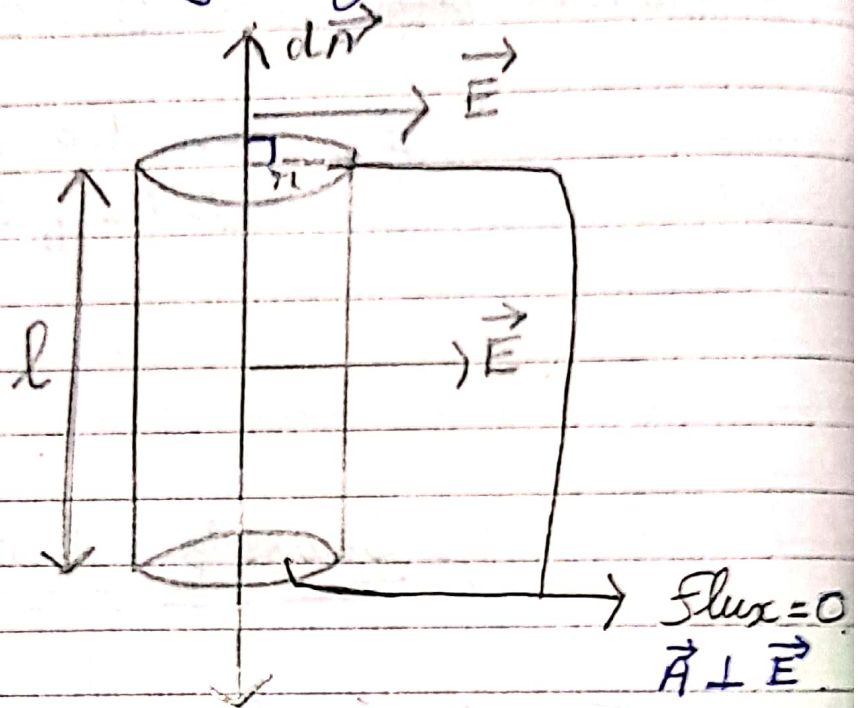
$$\epsilon_0 \rightarrow 8.85 \times 10^{-12}$$

$\epsilon_0 \rightarrow$ permittivity of free space.

Applications of Gauss's law

Derivation :-

- i) Electric field due to uniformly charged infinitely long wire



$\lambda \Rightarrow$ Linear charge density

• Because It is infinitely long wire.

$$\phi = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

$$\Rightarrow 0 + 0 + (2\pi r_1 l) E(1) \rightarrow 0 = 0 \quad \cos(0) = 1$$

↓
surface area of cylinder

by $\phi = E \cdot A \cos \theta$

$$\Rightarrow (2\pi r l) E(1) = \frac{Q_e}{\epsilon_0}$$

$$\Rightarrow (2\pi r l) E(1) = \frac{\lambda l}{\epsilon_0}$$

$$\Rightarrow E = \frac{\lambda}{2\pi r \epsilon_0} \times 2$$

$$\Rightarrow E = \frac{2\lambda}{\pi r \epsilon_0}$$

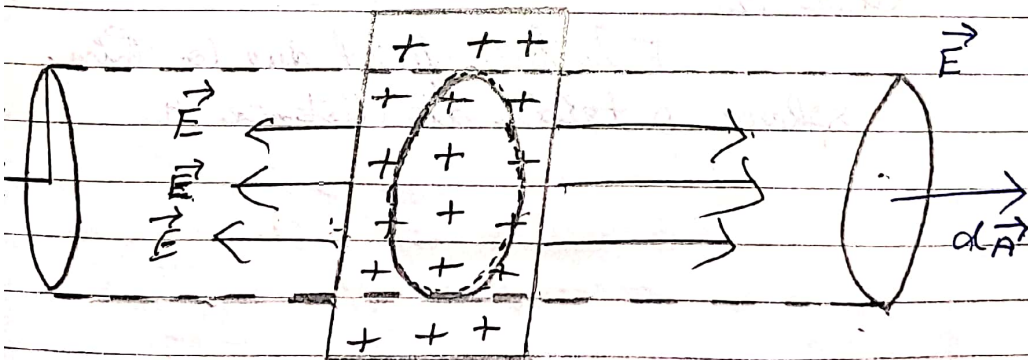
$$E = \frac{2K\lambda}{r}$$

$$E \propto \lambda$$

$$E \propto \frac{1}{r}$$

$$E = \frac{2K\lambda}{r}$$

ii) Electric field due to uniformly charged infinite plane sheet.



Surface Area flux = 0.

$$\Phi = EA + EA = 2EA = \frac{\sigma A}{\epsilon_0}$$

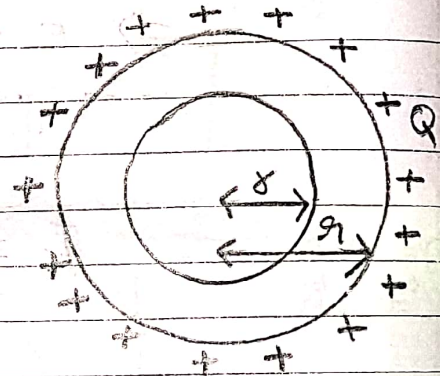
$$E = \frac{\sigma}{2\epsilon_0}$$

iii) Electric field due to Thin spherical ~~sph~~ shell :

Case : 1st.

Electric field at a point p inside the sphere at a distance r ($r < R$).

$$\phi = \frac{Q_{en}}{\epsilon_0}$$



$$E = 4\pi r^2 = \frac{0}{\epsilon_0}$$

$$E = 0$$

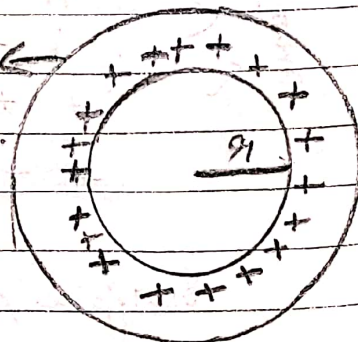
$$E 4\pi r^2 = E \int dA$$

both are same.

Case : 2.

Electric field due to Thin sphere outside at distance r .

Gaussian Surface.



$$\phi = \frac{Q_{en}}{\epsilon_0}$$

$$E = 4\pi r^2 = \frac{Q}{\epsilon_0}$$

$$E = \frac{1}{4\pi r^2 \epsilon_0} Q = \frac{kQ}{r^2}$$

$$E = \frac{kQ}{r^2}$$